Examining the effects of noise pollution on bat activity

Context

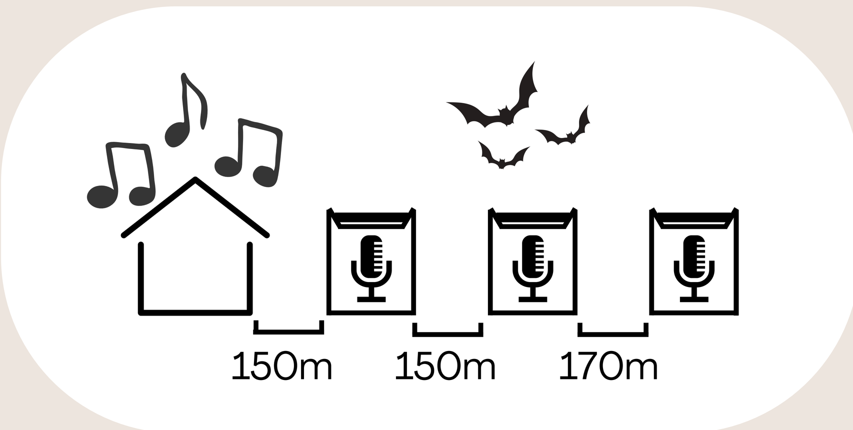
Bats, considering their wide range of environmental niches, are excellent **bioindicators**. Unfortunately, they are now under threat, particularly as a result of global climate change and human activity. **Light pollution** is heavily discussed when talking about bats, unlike **noise pollution**.

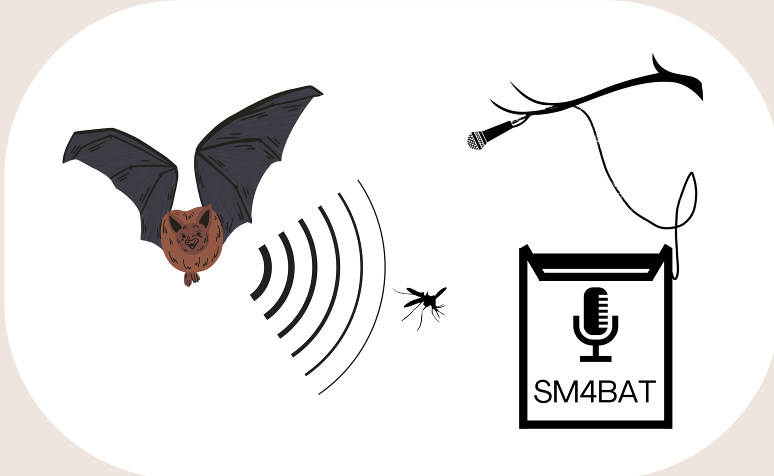
Material and method

The study was conducted in two areas :
- Le Domaine d'O - Montpellier
- Association Rising Dead Boys - La-Fare-Les-Oliviers

These areas are home to a good **diversity** of bats, and **noise-heavy** shows are regularly organized there. Recordings were done from **late september** to **early november**.

3 ultrasonic recorders were placed near noise-heavy areas. Recordings were made from **dusk** to **dawn** on the day before, day of and day after shows.





Data processing : We took into account three parameters: the effect of distance, the total number of detections per hour and night. Averages were taken into account for all nights and both locations. The graphs (ggplot function) allowed to compare the activity of the bats between nights.

Results

The results of our study reveal that **distance**, **time**, and interactions between **nights** and **distance** significantly influence the number of bats detected.

Furthermore, the nights following disturbances show an increase of approximately 55.2% in the number of **individual detections** ($p = 0.050$, figure 1). Nights with noise disturbance show a trend toward a **decrease** in the number of bat detections compared to nights without disturbance, though this effect remains marginally significant ($p = 0.064$).

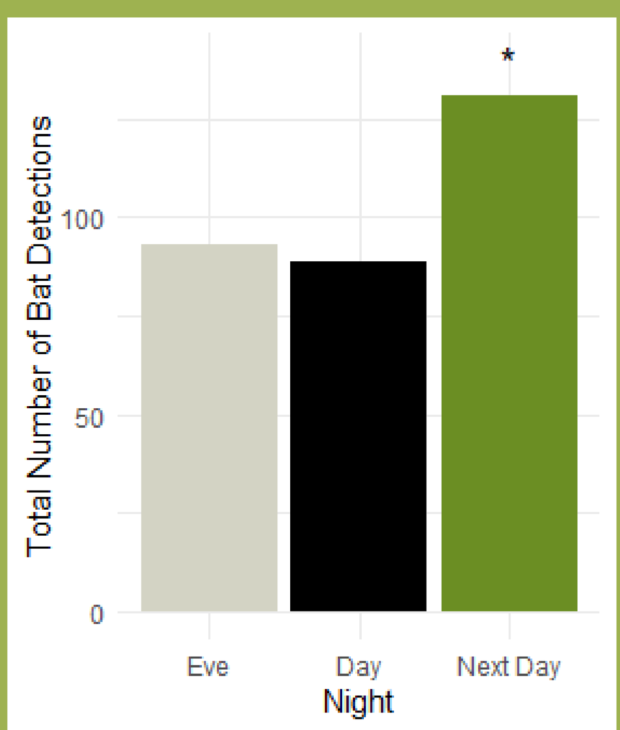


Figure 1: Number of bats detected depending on recording night... * = Statistical significance

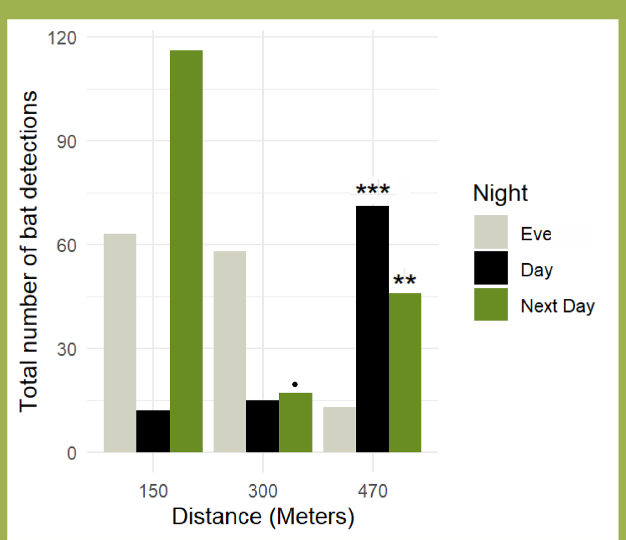


Figure 2: ...with distance variable
**** = Strong statistical significance**
***** = Very strong statistical significance**
• = Weakly significant

However, at 470 meters, there is a **significant increase** in the number of detections during disturbed nights (+295%, $p < 0.001$) and the following nights (+124%, $p = 0.003$).

Discussion

The **decrease** in bat activity might result from sound **interference** with echolocation. However, the limited effect could also stem from the short duration or low intensity of the **disturbance**. Alternatively, it is possible that bats are not significantly **affected** by noise pollution on concert days.

The **rise** in activity the days **following shows** could be due to two factors:

an abundance of **insects** attracted by the show

reduced **human activity** on Sundays

Additionally, concert noise appeared to **significantly increase** bat activity in **distant areas**, likely due to bats avoiding noise frequencies that **overlap** with their echolocation. This elevated activity persisted the **following night**, suggesting that bats temporarily **avoid** disturbed areas perceived as **inefficient** or **risky** for foraging.

Improvements

Recording over a **longer period**, such as the entire summer with 24-hour monitoring, and including more days and comparable locations, would have improved the **reliability** and **depth** of the data.

Take home message

Noise pollution disrupts bat's **echolocation**, a crucial mechanism for hunting and navigation. It limits their access to habitats and reduces food availability, thereby altering their **behavior** and potentially affecting **survival** and **reproduction**. Mitigating anthropogenic noise is essential to **safeguard** these vital species and maintain ecosystem balance.

FIRE Formation Innovante par la REcherche
UNIVERSITÉ PERPIGNAN VIA DOMITIA
UNIVERSITÉ MONTPELLIER
FdS FACULTÉ DES SCIENCES MONTPELLIER

Alexandre Constant
Edouard Distin
Clarisse Launay
Gabrielle Metz
Idriss Ramky